

Business Intelligence

26FS

Programme	BSc in Business Information Technology
Degree	Bachelor
ECTS	5
Module type	mandatory module
Module coordinator	Dr. Gwendolin Wilke
Compulsory attendance	Attendance is expected.
Leading principle / Short description	The first part of the module "Data Warehousing" (50%) provides an introduction to Business Intelligence (BI), with a focus on the fundamental concepts and technologies of Data Warehousing (DWH). In particular, we will discuss basic aspects of modeling multidimensional data structures and OLAP data analysis. The second part, "Introduction to Machine Learning" (25%), provides a first insight into the world of advanced analytics. Some of the fundamental concepts of machine learning and AI are introduced, and basic examples of supervised and unsupervised learning algorithms are discussed. The third part, "Data Storytelling" (25%) introduces the fundamentals and principles of visual information design and data storytelling. Students learn how complex data structures and the results of data analytics can be communicated in a target oriented and context aware fashion that supports managerial decision making.
Module content	◦ Topic 1: Data Warehousing (50%) ◦ Topic 2: Machine Learning (25%) ◦ Topic 3: Data Storytelling (25%)
Competencies to be achieved	Knowledge and Understanding: <i>Students know ...</i> ◦ fundamental concepts of BI (Business Intelligence) and DWH (Data Warehousing) ◦ characteristics and purpose of analytical information systems ◦ central concepts of semantic and logical multidimensional data modeling ◦ central concepts of the ETL (Extract-Transform-Load) process ◦ central aspects of data quality ◦ some fundamental concepts of machine learning ◦ the importance of data communication ◦ the basic methods and tools of data visualization and data storytelling ◦ some relevant cognitive biases Applying Knowledge and Understanding: <i>Students can ...</i> ◦ create a semantic model of a simple data set in the visual modelling language ADAPT ◦ create a logical model of a simple data set as a star schema ◦ perform OLAP (Online Analytical Processing) operations on a simple multidimensional data cube ◦ identify data quality issues in a simple data set ◦ identify and apply ETL (Extract, Transform, Load) transformations to a simple data set ◦ apply selected machine learning algorithms to simple datasets and correctly interpret the output. ◦ recognize and apply suitable types of visualizations and fundamental concepts of visual analytics ◦ apply principles of visual storytelling to a specific business case Making Judgment: The students can... ◦ distinguish between operational and analytical information needs ◦ differentiate levels of data modeling ◦ identify and categorize poor quality data ◦ evaluate the performance of selected machine learning algorithms ◦ assess whether selected machine learning methods can solve a given business problem ◦ distinguish misleading from correct visualizations ◦ assess the influence of visualization elements and narratives on decisions Communicative Skills: The students can... ◦ design visual dashboards and data stories
Prerequisites	Recommendations: ◦ Business Math 2 ◦ Database Technologies
Teaching and learning methods	◦ Contact Hours: Lecture, Discussion, Individual Exercises, Group work, Case Studies ◦ Guided Self-Study: Individual Exercises, Group work

Literature
Grading
Assessment

Required Reading: None

Grade 1-6 (half grades)

Exam 100%

Oral / Written

Written

Duration (min)

90

Timeframe

Examination period (end of semester)

Grading Scale

points

Remarks

The type of assessment may take a different form in the event of a repeat exam.

Module details**Business Intelligence (BITFT 2a) - Wed - Basel****Time** 8:15 AM - 12:00 PM**Language** Englisch**Max. participants** 64**Periodicity** Weekly**Lecturers** Dr. Gwendolin Wilke, Dr. Terry Inglese, Jacqueline Vitacco**Number** 2-26FS.W-B-BITPT-040406.SN/FT24**Business Intelligence (BITPT23) - Thu - Windisch****Time** 1:15 PM - 5:00 PM**Language** Englisch**Periodicity** Weekly**Lecturers** Dr. Gwendolin Wilke, Jacqueline Vitacco**Number** 1-26FS.W-B-BIT-050406S1.SN/PT23

